



KING KHALID UNIVERSITY ■ College of Computer Science

Genetic Mutation Detection Using Machine Learning

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Introduction

Problem statement: missense variants at clinical scale

Missense variants at clinical scale

- A point mutation that swaps one amino acid in a protein.
- Most common coding variant in the human genome.
- Most are benign; a clinically-important minority cause disease.
- Distinguishing the two is the central problem of clinical genetics.

Why prior work overstates performance

1. **Definitional circularity** — features derived from the label (e.g. `is_common`) inflate metrics.
2. **Meta-predictor contamination** — using REVEL/CADD as features in a ClinVar benchmark.
3. **Paralog leakage** — KRT1 in train + KRT14 in test $\Rightarrow$ 80% sequence identity $\Rightarrow$ model has seen most of the test.

Exome: $\sim 20,000$ variants/patient; human curation $\sim$ a few hundred/day. *Scale demands automation; automation demands honest evaluation.*

Research objectives + scope

Data + methodology

1. Assemble a paralog-disjoint corpus of 195k missense variants.
2. Eliminate three specific leakage sources.
3. Train and calibrate a defensible baseline.
4. Re-score published tools on our own test split.

Validation + delivery

5. Evaluate externally on DENOVO-DB.
6. Prove orthogonality of zero-shot ESM-2 signal.
7. Publish everything: code, tests, Docker, 10 narrative notebooks, Streamlit demo, academic report.

Every objective maps to at least one CI-enforced check.

Related works

Classical tools (2003–2010): conservation-based predictors

Tool (year)	Algorithm	Pros	Cons
SIFT (2003)	Conservation only — PSSM from PSI-BLAST	Instant via VEP REST; baseline of >95% clinical tools	No AA chemistry; arbitrary 0.05 threshold; weak on slow-evolving sites
PolyPhen-2 (2010)	Naïve Bayes on conservation + chemistry + 3D struct. (HumDiv)	Biochemical context (Grantham, BLOSUM); standard in clinical reports	HumDiv overlaps ClinVar $\Rightarrow$ contamination on ClinVar benchmarks

Where we re-score them

On our paralog-disjoint test (own-coverage): SIFT = 0.620 PR-AUC (96 % cov.), PolyPhen-2 = 0.728 (93 % cov.). See Section 9.

Gradient-boosted ensembles (2016–2024): supervised meta-predictors

Tool (year)	Algorithm	Pros	Cons
REVEL (2016)	Random forest ensemble of 13 prediction tools	Very high reported AUC (~ 0.95); widely adopted	Inputs ClinVar-trained $\Rightarrow$ circular when benchmarked on ClinVar
CADD (2019)	Logistic/RF on simulated negatives + observed positives	Single model for all variant types (incl. non-coding)	Scores uncalibrated; training labels are simulated
VARITY (2021)	XGBoost with noise-weighted training	Handles label noise explicitly	Gene-level split only (no paralog protection)
MAGPIE (2024)	Gradient boosting on multi-omics features	Recent, multi-modal feature space	Paralog-split status undocumented in the publication

Why we ban these as features

`verify_no_leakage.py` excludes REVEL, CADD, ClinPred, M-CAP, MetaSVM — using meta-predictors trained on ClinVar would inflate PR-AUC by ~ 0.07 .

Protein language models (2021–2023): unsupervised representations

Tool (year)	Algorithm	Pros	Cons
EVE (2021)	Variational auto-encoder, one per gene ($\sim 3,000$ genes)	Gene-specific deep model; captures epistasis	Limited gene coverage; slow per-gene training
ESM-1b / ESM-2 (2023, zero-shot)	Masked-AA log-likelihood ratio from 250M-sequence corpus	Fully unsupervised $\Rightarrow$ uncontaminated by ClinVar	Lower AUC at small scale (we use esm2_t12_35M)
AlphaMissense (2023)	Structural PLM calibrated against ClinVar at release	SOTA reported AUC (~ 0.96 on own benchmark)	Calibrated on ClinVar $\Rightarrow$ comparison contamination on ClinVar tests

The PLM choice in our pipeline

ESM-2 is the only one provably unsupervised $\Rightarrow$ we integrate its LLR as a Phase-2.1 feature (Section 8). Lift on internal test: $\Delta \text{PR-AUC} = +0.0313$, $p < 10^{-4}$.

Contribution

Our contribution + technology stack

Six contributions

1. **Leakage audit** — PR-AUC 0.955 $\rightarrow$ 0.836 raw / 0.827 calibrated, three named contamination sources removed.
2. **Like-for-like baseline re-scoring** — SIFT, POLYPHEN-2, ALPHAMISSENSE scored on *our own* paralog-disjoint test (own-coverage CIs).
3. **External validation on denovo-db** — paired bootstrap on Δ ROC (+0.076, $p = 0.073$) post-constraint vs. pre-constraint.
4. **ESM-2 dual integration** — Phase 1 zero-shot rank fusion + Phase 2.1 training-time feature (Δ PR-AUC +0.0313, $p < 10^{-4}$).
5. **Auditable methodology** — 95 % bootstrap CI on every metric, Murphy decomposition, paired tests.
6. **Reproducible artifact** — 157 tests, CI-enforced leakage gate, Docker image, Streamlit demo, LaTeX academic report.

github.com/RayanAlDwlah/Genetic-Mutation-Detection-project

Technology stack

ML core: XGBOOST 2.x, OPTUNA 4 (TPE, 40 trials), SHAP, scikit-learn

Protein language model: HuggingFace transformers, esm2_t12_35M_UR50D (35M params)

Data + features: CLINVAR, GNOMAD v2.1 + v4, DBNSFP 5.3.1a; pandas, polars, parquet/pyarrow

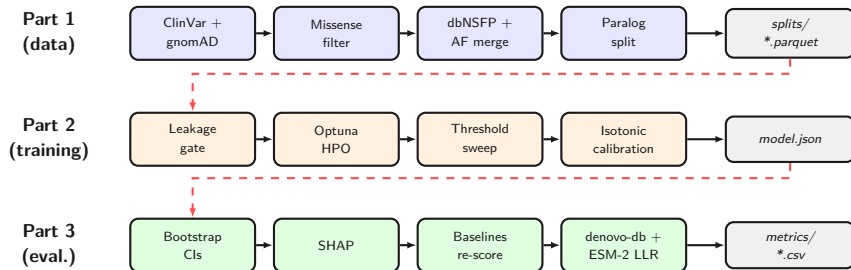
External APIs: ENSEMBL VEP REST (with on-disk cache fallback)

Infrastructure: Docker (TeX Live 2026), GitHub Actions CI, pytest 157, ruff, black, mypy, pre-commit

Delivery surfaces: STREAMLIT demo, TYPER CLI, 10 narrative Jupyter notebooks, LaTeX defense + thesis

Proposed System

Pipeline overview — three parts, three lanes



Dashed red arrows = inter-part hand-offs (Part 1 splits feed Part 2; Part 2 model feeds Part 3).

Four datasets joined on canonical chr:pos:ref:alt (GRCh38)

- **ClinVar** — pathogenic / benign labels, review star ≥ 2 ; $\sim 283,392$ candidate variants.
- **gnomAD v2.1 + v4** — allele frequency, allele count, allele number.
- **dbNSFP 5.3.1a** — conservation (phyloP, GERP++), AA chemistry (Grantham, BLOSUM62), Pfam domain annotations.
- **gnomAD v2.1.1 constraint** — gene-level pLI, LOEUF, mis_z, oe_mis_upper, lof_z.

Output to Part 2

splits/{train,val,test}.parquet after the two filters described in Part 1.2 + 1.3.

The single biggest audit finding

Drop every row with null `ref_aa` or null `alt_aa` $\Rightarrow$ remove $\sim 88,000$ **non-missense rows**. Result: 195,098 **true missense variants** survive.

- **Pre-audit:** 64% of “pathogenic” rows had null AA fields — the model was learning “null AA $\Rightarrow$ pathogenic”, *not biology*.
- **Fix alone drops PR-AUC 0.955 $\rightarrow$ 0.819 — the largest single drop in the entire audit.**
- Implemented in `src/clinvar_cleaning.py`; CI-enforced via `verify_no_leakage.py` check #2.

Why a plain gene-level split fails

52% of prefix families (ZNF*, SLC*, KRT*, TMEM*) are shared between train and test under a naïve gene split. With $\geq 80\%$ sequence identity between paralogs, the model has effectively seen the test set.

Fix: family-level GroupShuffleSplit via `assign_gene_family` —

- 16 regex patterns (KRT*, ZNF*, etc.) + trailing-digit fallback.
- 15,479 **genes** $\rightarrow$ 7,851 **families**.
- Zero family overlap across train / val / test, CI-enforced.

Motivated by Heijl et al. (2020): 5–15 percentage-point ROC inflation under random splits, measured across 55 published predictors.

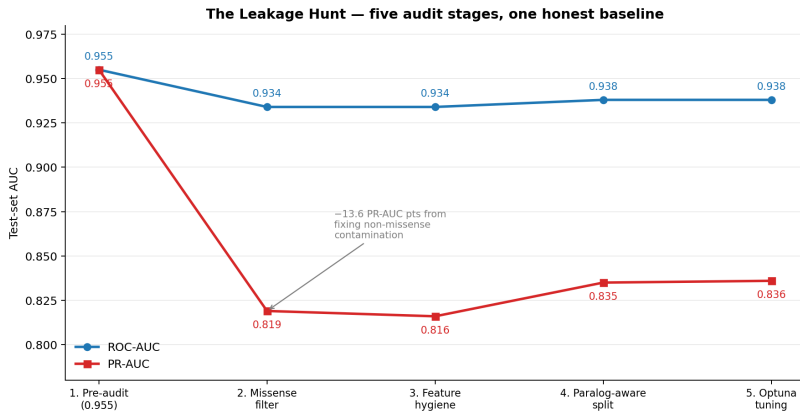
Part 2.1 — Leakage gate (6 CI-enforced invariants)

1. **Banned features** — no `is_common`, `chr`, `ref`, `alt` in the training matrix; no ClinVar-trained meta-predictors (REVEL, CADD, ClinPred, M-CAP, MetaSVM).
2. 100% **missense-only** rows in the training split (Part 1.2 filter applied).
3. **Split disjointness** — zero shared genes *and* zero shared paralog families across train / val / test.
4. **Label-rate gap** < 0.08 across splits (prevents stratification leakage through prevalence drift).
5. **Phase-2.1 ESM-2 features** pass banned-upstream-meta check (LLR not derived from any ClinVar-trained scorer).
6. **ESM-2 split integrity** — no test/val overlap in the score parquet returned from the Colab batch.

Every push to main runs all six

Failure $\Rightarrow$ blocks the merge via GitHub Actions. Source: `src/verify_no_leakage.py:CHECKS`.

Part 2.2 — Five-stage leakage journey



Raw PR-AUC across the audit: 0.955 → 0.819 → 0.816 → 0.835 → 0.836.

Stage 2 (missense filter) is the largest single drop; stages 3–5 recover modestly through feature hygiene + paralog split + Optuna re-tune.

Calibrated headline: 0.827 [0.819, 0.835].

Part 2.3 — Model architecture + hyperparameter optimisation

Model: XGBoost

- Gradient-boosted decision trees (XGBoost 2.x).
- 78 features post-hygiene (Phase 1).
- 80 features after ESM-2 LLR addition (Phase 2.1).
- Train: 138,921 variants, val: 28,079.

Optuna TPE search

- 40 trials, multivariate grouping.
- Objective: PR-AUC on validation.
- Median pruner after 200 rounds.
- Early stopping = 80 boost rounds.
- Fixed seed = 42.

Result: a flat hyperparameter landscape

Top-10 trials within 0.001 PR-AUC of each other. The baseline ceiling is real — further gains require better *representations* (ESM-2 LLR, AlphaFold structure), not more search.

Part 2.4 — Isotonic calibration + Murphy decomposition

$$\text{Brier} = \text{Reliability} - \text{Resolution} + \text{Uncertainty}$$

Calibrator	Brier	Reliability	Resolution	ECE	Log-loss
Raw	0.0876	0.00543	0.1055	0.054	0.286
Platt	0.0834	0.00114	0.1055	0.029	0.278
Isotonic	0.0826	0.00024	0.1053	0.011	0.273

- Resolution constant $\Rightarrow$ discrimination unchanged (isotonic is a monotone post-hoc transform).
- Reliability drops **23 $\times$** $\Rightarrow$ probability calibration works.
- ECE $0.054 \rightarrow 0.011 < 0.02$ *clinical target*.

Paralog-disjoint test split

28,098 **variants**, fully disjoint from train + val at the paralog-family level. ~30% pathogenic class rate.

Statistical methodology

- 95% **bootstrap CI** on every reported metric — ROC-AUC, PR-AUC, F1, MCC, Brier — 1,000 replicates, seed = 42.
- **Murphy decomposition** for calibration: separates discrimination from probability quality (Part 2.4).
- **Paired bootstrap on Δ** for ablations — *same resampled indices* for both compared models, $n = 28,098$, gives a much tighter CI than two independent runs.

No single-number claims; every metric reported with an interval.

denovo-db family-holdout slice

$n = 201$ variants in gene families completely unseen during training. Cohort: affected vs control de-novo missense.

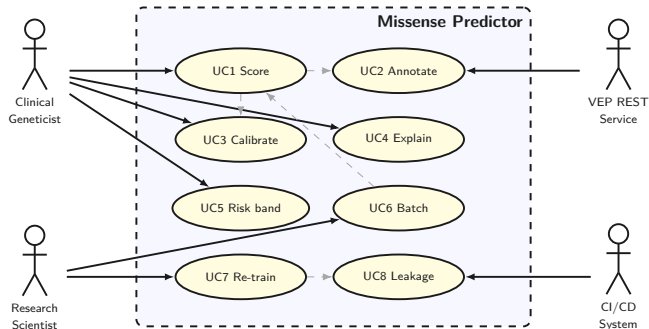
Three orthogonal tests

- **Permutation test** for $H_0: \text{AUC} \leq 0.5$ (10,000 replicates, seed = 42).
- **Paired bootstrap on $\Delta = \text{AUC}_{\text{post}} - \text{AUC}_{\text{pre}}$** (constraint features), recovered pre-constraint checkpoint at f8ab464.
- **Calibration transfer audit** — does isotonic from ClinVar val ($\sim 24\%$ pathogenic) transfer to denovo-db's $\sim 80\%$ prior? Spoiler: it does *not* (D3).

External validation guards against ClinVar-shaped overfit.

System Analysis & Design

Use case diagram + requirements summary



Functional (×8)

Ingestion, feature assembly, scoring, risk categorisation, feature attribution, batch mode, external reproduction, report regeneration.

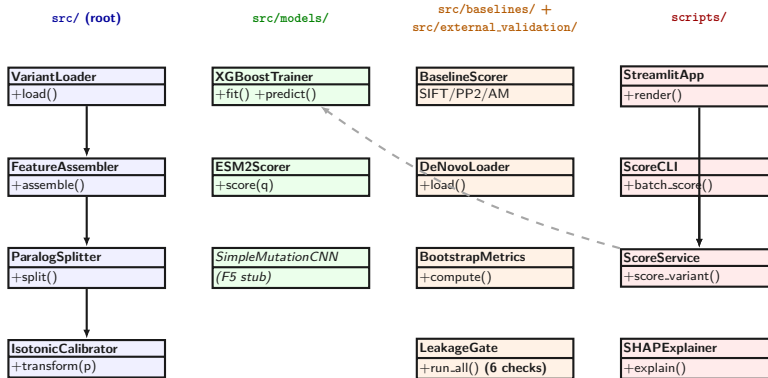
Non-functional (×7)

Performance (<100 ms cache hit), reliability (10^{-3} seed-reproducible), scalability (10^6 variants), usability (non-ML user), maintainability (CI), security (no PHI), reproducibility (Docker).

Traceability

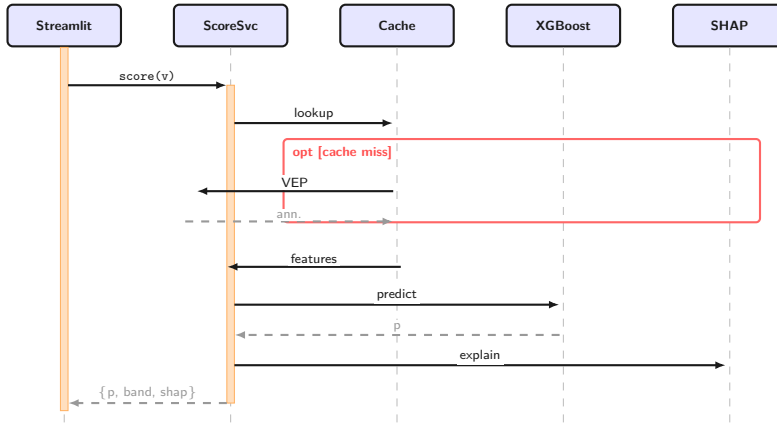
Every requirement maps to ≥ 1 src/ module + ≥ 1 pytest test case.

Class diagram — four *real* packages from `src/`



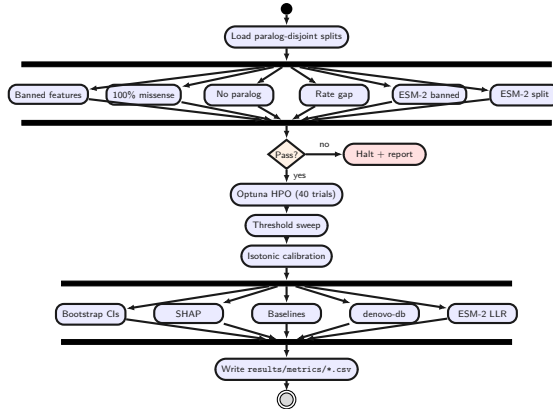
Conceptual blueprint. Production code uses module-level functions in `src/*` rather than the named classes shown here, following Python convention for ML pipelines.

Sequence diagram — single scoring request



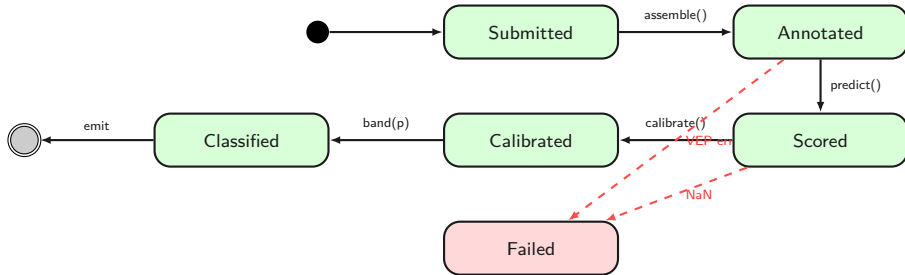
Conceptual blueprint. ScoreSvc, Cache, XGBoost, SHAP map to module-level functions in `scripts/streamlit_app.py` + Pickle/Parquet I/O, not concrete classes.

Activity diagram — training + evaluation workflow



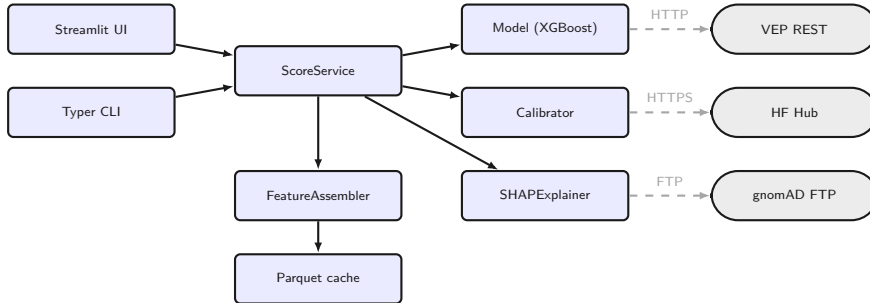
Fork-1 = 6 leakage checks (parallel); Fork-2 = 5 eval tracks. Conceptual flow; production uses sequential scripts.

State diagram — variant life-cycle



Five-state happy path (Submitted → Annotated → Scored → Calibrated → Classified). Two error transitions to Failed: upstream VEP REST errors and downstream NaN feature errors.

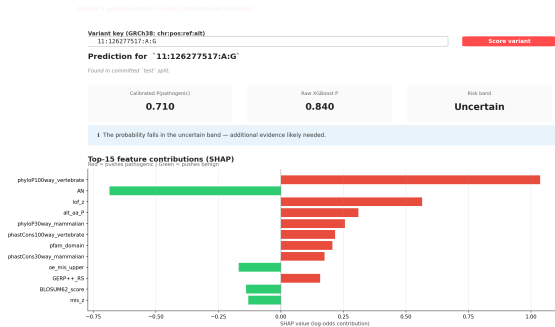
Component diagram — module-level dependencies



*Solid arrows = direct uses; dashed grey = external network dependencies (VEP HTTP, HF Hub HTTPS, gnomAD FTP).
Conceptual blueprint; production uses module-level functions, not concrete classes.*

Interface

Interface — Streamlit demo + Typer CLI



Streamlit web demo

Input GRCh38 variant key chr:pos:ref:alt
Output calibrated p + raw XGBoost + risk band
Explainer Top-15 SHAP, colour-coded by direction
Latency <100 ms cache hit; ~2s cache miss

Typer CLI

Single score chr:pos:ref:alt
Batch score-batch <vcf|csv>
Explain explain --top 15 <variant>
Re-train make train (Optuna 40 trials)

Implementation and experimental results

Implementation — hardware, software stack, reproducibility

Hardware

- Apple Silicon laptop (MPS GPU)
- Free-tier Colab T4 (ESM-2 scoring only)
- GitHub Actions ubuntu-22.04

Software stack

- Python 3.11 + pinned deps
- XGBOOST 2.x + OPTUNA 4
- SHAP, scikit-learn, scipy
- HF transformers, PyTorch
- esm2_t12_35M_UR50D
- pandas, polars, parquet, pyarrow

Reproducibility

- Docker (TeX Live 2026)
- requirements-lock.txt
- Fixed seeds (42)
- **157 pytest tests**, 59% coverage
- CI: ruff + black + mypy + leakage gate
- Snapshot tests on metric files

Single-command reproduction

```
docker build -t missense-classifier . &&  
docker run --rm missense-classifier make reproduce-headline
```

Reproduces every metric within 10^{-3} on a clean clone.

Headline results — paralog-disjoint test split

Test split

Paralog-disjoint

28,098 variants

Pathogenic rate $\sim 30\%$

All 95 % CIs from 1,000-replicate nonparametric bootstrap (seed = 42).

Metric	Value (95 % CI)
ROC-AUC	0.938 [0.935, 0.941]
PR-AUC (uncalibrated)	0.836 [0.827, 0.843]
PR-AUC (calibrated)	0.827 [0.819, 0.835]
F1	0.775
Brier (calibrated)	0.083
ECE (calibrated)	0.011

Calibration trades a small ranking penalty ($\Delta\text{PR-AUC} = -0.008$, $0.8355 \rightarrow 0.8273$) for $\sim 5\times$ better probability reliability (ECE $0.054 \rightarrow 0.011$). Calibrated probabilities are the headline; uncalibrated is shown for transparency.

Highlighted rows: holdout from gene families completely unseen in training.

Slice	n	ROC-AUC (95 % CI)	PR-AUC (95 % CI)
full (pre-constraint)	642	0.468 [0.42, 0.52]	0.761
full (post-constraint)	642	0.511 [0.46, 0.56]	0.790
holdout (pre-constraint)	201	0.487 [0.38, 0.58]	0.789
holdout (post-constraint)	201	0.573 [0.48, 0.67]	0.838

The headline

On family-holdout ($n = 201$): paired bootstrap $\Delta\text{ROC} = +0.076$ $[-0.021, +0.172]$, one-sided $p = 0.073$. **Partial closure** of the generalisation gap, not a deployable predictor.

C1 — near-base-rate caveat

Holdout class balance is **80 % positive**.
Constant-positive baseline PR-AUC ≈ 0.80 .
Meaningful PR-AUC lift is only $+0.037$, so ROC-AUC is the primary metric for this cohort.

ESM-2 zero-shot + rank fusion

Test split (28,098 variants, paralog-disjoint)

Model	ROC-AUC	PR-AUC	α
XGBoost (calibrated)	0.938	0.827	—
ESM-2 LLR (zero-shot)	0.735	0.604	—
Rank fusion, uniform	0.895	0.784	0.50
Rank fusion, tuned	0.941	0.852	0.175

$$s_i^{\text{fused}} = (1 - \alpha) \cdot \frac{r_i^{\text{XGB}}}{n} + \alpha \cdot \frac{r_i^{\text{ESM}}}{n}, \quad \alpha^* = 0.175$$

Why fusion works

Tuned $\alpha = 0.175$ down-weights the weaker PLM signal.

Uniform $\alpha = 0.5$ is *worse* than XGBoost alone. Tuned mix beats XGBoost on internal test (paired DeLong on denovo-db: n.s. at the 5% level).

Phase 1 takeaway

ESM-2 zero-shot LLR is fully unsupervised — provably uncontaminated by ClinVar. Fusion gives +0.025 PR-AUC lift over XGBoost, motivating training-time integration (Slide 21).

Phase 2.1 — training-time ESM-2 integration + SHAP shift

Ablation (paralog-disjoint test, $n = 28,098$)

Model	ROC	PR
Phase-1 baseline	0.938	0.827
Augmented no_esm2	0.939	0.840
Augmented esm2_only	0.739	0.610
Augmented full	0.948	0.865

Paired bootstrap full vs. no_esm2:

$\Delta\text{PR-AUC} = +0.0313$ [$+0.0266, +0.0361$], $p < 10^{-4}$.

$\Delta\text{ROC-AUC} = +0.0093$ [$+0.0082, +0.0103$], $p < 10^{-4}$.

SHAP shift after ESM-2 added

- `is_imputed_esm2_llr` ranks #1 (mean $|\text{SHAP}| = 1.05$)
- `esm2_llr` ranks #4 ($= 0.42$)
- Conservation features (phyloP, GERP) remain top-tier; ESM-2 partly substitutes (Spearman $\rho \leq |0.31|$), partly supplements.
- `lof_z` drops from rank 3 (Phase-1) to rank 5.

denovo-db holdout: $\Delta\text{ROC} = -0.13$ + C2 constraint-coverage caveat

The in-distribution gain ($+0.031$ PR-AUC) does *not* transfer to unseen families. **Partly a coverage artefact (C2):** held-out family genes receive *imputed* constraint values for the augmented model where the baseline had real gnomAD values. Full slice (preserved coverage) shows no degradation — supporting the artefact interpretation, not a pure ESM-2 effect.

Comparison with published methods

All baselines re-scored on our paralog-disjoint test split (28,098 variants).

Method	Year	Methodology	Coverage	PR-AUC
SIFT	2003	legacy, no guards	96%	0.620
PolyPhen-2 [†]	2010	legacy, no guards	93%	0.728
AlphaMissense [*]	2023	high ClinVar risk	86%	0.890
ESM-2 LLR (zero-shot)	2023	no supervision	100%	0.604
XGBoost (ours, baseline)	2026	paralog + audit	100%	0.827
XGBoost + ESM-2 LLR (ours)	2026	paralog + audit	100%	0.865

What only we report

- Paralog-family-disjoint split (CI-enforced)
- Three-source leakage audit
- External validation on DENOVO-DB
- Bootstrap 95 % CIs + paired bootstrap on Δ
- Calibration deep-dive (ECE = 0.011)
- Docker + lock file: 6-decimal reproduction

Honest framing

- +0.245 PR-AUC over SIFT
- +0.137 PR-AUC over PolyPhen-2
- AM gap: -0.063 (Phase-1) $\rightarrow$ **-0.025** (Phase-2.1)
- AM is ClinVar-calibrated; matched-coverage on $\mathcal{S}_{AM}$ shrinks gap further (D7).

Each row on its baseline's own-coverage subset — rows NOT directly comparable.

^{*} AM ClinVar-calibrated $\Rightarrow$ test overlaps training signal. [†] PolyPhen-2 trained on HumDiv/HumVar (overlaps ClinVar). REVEL omitted (13 ClinVar-trained base learners).

Conclusion and future works

Conclusion — take-home messages

1. **Audited numbers beat inflated ones.** Calibrated PR-AUC 0.827 has no known contamination; the pre-audit 0.955 had three.
2. **Like-for-like baselines change the conversation.** On the same paralog-disjoint test we beat classical tools cleanly (SIFT 0.620, PolyPhen-2 0.728) and sit just below AlphaMissense (-0.025 PR-AUC) under stricter methodology.
3. **Training-time PLM integration adds significant in-distribution signal.** The ESM-2-augmented model lifts test PR-AUC $0.827 \rightarrow 0.865$ ($\Delta = +0.031$, $p < 10^{-4}$, paired bootstrap on $n = 28,098$).
4. **External validation is the real benchmark.** Partial closure on DENOVO-DB (paired $\Delta\text{ROC} = +0.076$, CI still covers chance at $n = 201$); the in-distribution ESM-2 gain does *not* transfer to unseen gene families ($\Delta\text{ROC} = -0.13$, motivates a larger PLM checkpoint).

1. **Larger-checkpoint ESM-2.** Upgrade to esm2_t33_650M_UR50D ($19\times$ parameters); expected to lift the OOD ceiling (current 35M shows $\Delta\text{ROC} = -0.13$ on unseen gene families).
2. **AlphaFold-2 structural features.** pLDDT, SASA, DSSP secondary structure, residue depth — orthogonal to PLM signal, complementary to conservation.
3. **ProteinGym DMS external validation.** Third dataset independent of ClinVar and DENOVO-DB; deep mutational scanning fitness measurements as both validation set and auxiliary training signal.
4. **Hybrid stacking + ACMG integration.** Meta-learner combining XGBoost + ESM-2 + AlphaMissense, packaged as an ACMG-compatible plug-in for clinical workflows.